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Salem

102208

QMS

QUARTERLY EXAMINATION- 2026

12 - Std

Time: 3.00 Hrs.

BIOLOGY

Marks : 70

BIO - BOTANY

I. Answer all the questions.

Choose the correct answer and write the option code. **(8x1=8)**

1. **Assertion:** Geitonogamy is a type of cross pollination.
Reason : In Geitonogamy, the pollen deposits on another flower of the another plant. a. Assertion and Reason are correct b. Assertion is correct. Reason is wrong c. Assertion is wrong. Reason is correct d. Assertion and Reason are wrong

2. Muller was the first to find out physical mutagen in
a. Maize b. Nicotiana c. Drosophila d. Oenothera

3. Southern blotting is
a. The transfer of DNA from agarose gels to nitrocellulose membrane.
b. The transfer of RNA to nitrocellulose membrane.
c. Electrophoretic transfer of Proteins to nitrocellulose membrane.
d. Attachment of probes to DNA fragments

4. Extra nuclear inheritance is a consequence of presence of genes in
a. Mitochondria and chloroplasts b. Endoplasmic reticulum and mitochondria
c. Ribosomes and chloroplast d. Lysosomes and ribosomes

5. "Gametes are never hybrid". This is a statement of
a. Law of dominance b. Law of independent assortment
c. Law of segregation d. Law of random fertilization

6. **Match List I with List II**

List I	List II
A. A pair of chromosomes extra with diploid	i) monosomy
B. One chromosome extra to the diploid	ii) tetrasomy
C. One chromosome loses from diploid	iii) trisomy
D. Two individual chromosomes lose from diploid	iv) double monosomy

a. A-i, B-iii, C-ii, D-iv b. A-ii, B-iii, C-iv, D-i
c. A-ii, B-iii, C-i, D-iv d. A-iii, B-ii, C-i, D-iv

7. The process of recombinant DNA technology has the following steps
I. Amplification of the gene II. Insertion of recombinant DNA into the host cells.
III. Cutting of DNA at specific location using restriction enzyme.
IV. Isolation of genetic material (DNA)
Pick out the correct sequence of step for recombinant DNA technology.
a. II, III, IV, I b. IV, II, III, I c. I, II, III, IV d. IV, III, I, II

8. Select the incorrect statement from given statements
a. A tonic used for cardiac arrest is obtained from *Digitalis purpuria*
b. Medicine used to treat rheumatic pain he is extracted from *Capsicum annum*
c. An anti malarial drug is isolated from *Cinchona officinalis*.
d. Anti-carcinogenic property is not seen in *Catharanthus roseus*.

II. Answer any four of the following questions. **(4 X 2 = 8)**

9. What is test cross?
10. Name the chemicals used in gene transfer.
11. What is the difference between missense and nonsense mutation?
12. Draw and label the parts of first cell of male gametophyte.
13. What are embryoids?
14. Write short note on *Sharbati Sonora.*

III. Answer any three of the following questions.

Question No. 19 is Compulsory. **(3 X 3 = 9)**

15. Give an account on Cryopreservation.
16. List out the functions of tapetum
17. What do you know about the word pBR322.

second

Sign

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18. Write any three reasons for Mendel's success in his breeding experiment.

19. P – Parental Generation – $2n = 4x = 28$ (A) X $2n = 2x = 14$ (B)

G - Gametes

$2n = 3x = 21$ (C)

i. Write the gametic conditions of A and B. ii. Write the name of the plant C

IV. Answer the following questions in detail. (2 X 5 =10)

20. a) Describe Incomplete dominance with an example. **(OR)**
b) With a suitable diagram explain the structure of an ovule.

21. a) Explain the mechanism of crossing over. **(OR)**
b) Write the applications of plant tissue culture.

BIO-ZOOLOGY

I Choose the correct answers: 8x1=8

1. The mode of sexual reproduction in bacteria is by
a) Formation of gametes b) Endospore formation
c) Conjugation d) Zoospore formation

2. The most important hormone in initiating and maintaining lactation after birth is
a. Oestrogen b. FSH c. Prolactin d. Oxytocin

3. The most common substrate used in distilleries for the production of ethanol is_____
a) Soyameal b) Ground gram c) Molasses d) Corn meal

4. ZO-ZZ system of sex determination occurs in
a) Fishes b) Reptiles c) Butterfly d) All of these

5. An operon is a: a) Protein that suppresses gene expression
b) Protein that accelerates gene expression
c) Cluster of structural genes with related function
d) Gene that switched other genes on or off

6. Human beings and animals of the ancient city of Pompeii were preserved mainly due to....
a) Ocean salinity b) Glacial ice c) Volcanic ash d) Sedimentary deposits

7. AIDS virus has a) Single stranded RNA b) Double stranded RNA
c) Single stranded DNA d) Double stranded DNA

8. Mammograms are generally recommended for women above the age of---
A) 70 years B) 35 years C) 40 years D) 45 years

II Answers any four of the following questions: 4X2=8

9. How is polyspermy avoided in humans?
10. Define-Surrogacy.
11. What is multiple allelism? Give one example.
12. How is the two stage process of protein synthesis advantageous?
13. Write short notes about Atavistic organs.
14. Signs and symptoms of mental depression.

III. Answer any three of the following questions: (Q.No 19 is compulsorY)

15. How is milk converted into curd? Explain the process of curd formation. 3X3=9
16. Difference between seasonal breeders and continuous breeders.
17. Write short notes on Kala – azaror *visceral leishmaniasis.*
18. What are interferons? Mention their role.
19. Why did Hershey and Chase use radioactively labelled phosphorous and sulphur only? Would they have got the same result if they use radiolabelled carbon and nitrogen?

IV. Answer all the questions: 2 X 5 = 10

20. a) Explain the role of oxytocin and relaxin in parturition and lactation. *(OR)*
b) Mention the main objections to Darwinism.

21. a) Differences between active and passive immunity. *(OR)*
b) It is established that RNA is the first genetic material? Justify giving reasons.

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